

Block Trades, Market Reaction and Monitoring Role: Evidence from India

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Abstract

We examine relationship between short term market reaction and diverse class of blockholders using a large sample of block trades from 2005–2015, in an emerging market context, India. We extend prior literature on broad typology of blockholders, and include insider blocks in addition to pressure-resistant and pressure-sensitive categories. We find abnormal returns on block trades by pressure-resistant and insider blocks is positive, compared to pressure-sensitive blocks. While existing literature highlights monitoring role of pressure-resistant blocks, we provide additional evidence of positive market reaction to insider blocks that signals promoter confidence and incentive to monitor management given large amount of wealth at stake. We find this effect to be strong in family owned firms, rather than family managed and non-family firms. Further, our results suggest that improved transparency and information disclosures pertaining to insider shareholding, impacts effective monitoring by pressure-resistant blockholders. Finally, we support prior evidence that extent of monitoring by pressure-resistant and insider blocks, is moderated by internal corporate governance mechanism as well as presence of large institutional investors, since they are considered as substitutes (not compliments) for each another.

JEL Classification: G20, G32, G34

Keywords: Blockholders, Block ownership identity, Corporate Governance

1. Introduction

This paper examines the relationship between short term market reaction and diverse class of blockholders using a large sample of block trades from 2005–2015, in an emerging market context, India. In a recent paper, Edmans and Holderness (2017) argue that blockholders are ubiquitous, however differences in institutional context and regulatory environments across countries, makes it interesting to study how different blockholders ameliorate or exacerbate their fundamental role of monitoring management. Role of blockholder governance in developed markets, characterized with dispersed ownership, arises from goal incongruence between shareholders and management. In contrast, governance challenges in emerging markets arises from unaddressed conflicts between the dominant blockholder and minority shareholders, manifested in concentrated ownership, extensive family ownership and control, business group structures, and weaker investor protection (La Porta et al., 2000).

In the Indian context, for instance, most often the dominant blockholder is an insider, evident in family owned and controlled firms, without much separation between ownership and control (Chakrabarti, Megginson and Yadav, 2008). The traditional view supports the entrenchment hypothesis (Morck et al., 1988) that family firms often control the firm and expropriate minority shareholders' interests (Dharwadkar et al., 2000; Morck et al., 2005). In contrast, the alignment of interest hypothesis postulates that higher family ownership results in greater alignment of interest between the controlling owners and other stakeholders (Demsetz and Lehn, 1985). As such, given a large amount of wealth at stake, these insider shareholders have the incentive to monitor actively, scrutinize management, and curb rent extraction (Shleifer and Vishny, 1986; Sarkar and Sarkar, 2000). Hence, a question for empirical investigation is whether blockholder governance matter in these firms in ways that are different in widely held firms.

Extant literature identifies broad typology of blockholders as *pressure-resistant* and *pressure-sensitive* groups¹ considering their economic incentives to monitor (Brickley, Lease, and Smith, 1988; Hutchinson, Seamer and Chapple, 2015; Muniandy, Tanewski, and Johl, 2016). The institutional setting of Indian firms, characterized by high promoter holding, provides a motivation to adequately identify and classify *insiders* (promoters, promoter group companies, promoter trusts, and executive/promoter directors) as a separate class of blockholders, because insiders can engage in monitoring as well as reap private benefits of control. Moreover, we focus on whether family firm dominance, manifested in ownership and control structures, affects governance role by diverse blockholder categories in such firms.

The prevalence of blockholder ownership raises a fundamental question regarding their role in effectively monitoring management. Khanna and Palepu (2000) argue that monitoring by large shareholders in developing countries may not be as effective as in developed countries because of poor availability of information due to inadequate disclosure norms and weak enforcement. In this context, reforms in the corporate governance regulation in India provides

¹Pressure-resistant category include investment firms and independent individual investors (includes HUF), that are in better position to actively monitor target firm due to their fiduciary responsibility and economic incentive to protect their investments. Corporations are recognised as pressure-sensitive category due to their conflict of interest with target firms emerging from current or potential business interests and product-market synergies (Brickley et al., 1988; Allen and Phillips, 2001; Fee, Hadlock, and Thomas, 2006).

a natural setting to test this hypothesis. More specifically, in 2009², disclosures about the ownership pattern specifically pertaining to insiders (promoter and promoter group) was amended, such as to bring more transparency and access to information regarding identity of different ownership groups on public domain, through quarterly financial reporting. Such an exogenous regulatory intervention provides the second motivation for the paper. We test the impact of this important legislative change, that aimed to improve greater information and transparency in disclosure norms, on scope of monitoring by blockholders during pre and post regulatory reforms periods.

Previous studies report that potential gains from added monitoring are greater when internal or external monitors are weaker (Shleifer and Vishny, 1986; Maug, 1998). Therefore, we investigate whether effective corporate governance mechanisms as internal monitors and presence of large institutional holding as external monitors, are substitutes (not compliments) to formation of outside blocks (Park, Selvili and Song, 2008).

In the context of defining blockholding, most countries use various threshold levels, such as 10%, 20% or more popularly the 5% ownership³ (including India) is justified, since it mandates public disclosure. Edmans and Holderness (2017) reason that theory offers little guidance for these threshold levels, rather ownership levels used to define a large blockholder appear to be relevant empirically. These levels could vary due to country-specific regulation, and firm specific characteristics. In this context, the SEBI regulation regarding Listing Agreement in 2001 required shareholding disclosure for more than 1% equity, and reporting the same in quarterly financial statements. Following Edmans and Holderness (2017), we explore the impact of these large-but-less-than-5% blocks, as external monitors.

Block trades⁴ are important stock market events and trades of majority blocks influence stock prices (Barclay and Holderness, 1991; Edmans, 2009). Studies show that market reacts differently to trades by diverse class of blockholders; Park et al., 2008 show that block trades by activist and strategic block purchases manifest as positive market reaction, compared to financial blocks. Chan et al. (1997) show positive market response to announcements of strategic alliances, and Barclay and Holderness (1991) find stock price increases when blockholders eventually acquire the firm. If blockholders do not impact cash flows that eventually accrues to minority shareholders, such stock price changes would be hard to explain (Barclay and Holderness, 1991, 1992). In our paper, we examine the short-term market reaction (3-day cumulative abnormal returns, CAR) to block trades as a function of blockholder categories (pressure-resistant, pressure-sensitive and insider blocks), family owned and managed firms, regulatory changes, prior institutional holding, and firm specific

² The Securities and Exchange Board of India (SEBI) vide its circular no. SEBI/CFD/DIL/LA/2009/3/2 dated February 03, 2009 issued amendments to Clause 35 and Clause 41 of the Equity Listing Agreement, regarding disclosures of shareholding of promoter and promoter group entities, and reporting the same in quarterly financial results of the company.

³ In the United States, this threshold is drawn from the legal definition of blockholders under Rule 13d-1(a) of the Securities Exchange Act of 1934. In India, the threshold of 5% and more of voting rights requires public disclosure under the Substantial Acquisition of Shares and Takeover Regulations 1997 (Takeover Code).

⁴ SEBI vide its circular no. MRD/DoP/SE/Cir-19/05 dated September 2, 2005 issued guidelines for defining block trade/block deal as, “A trade, with a minimum quantity of 5,00,000 shares or minimum value of Rs. 5 crores executed through a single transaction on a separate window of the stock exchange will constitute a “block deal”. However, while defining a large blockholder, we use the threshold of 5% ownership because it triggers substantial acquisition, and requires public disclosure.

characteristics, using possibly a large sample of block trades from 2005–2015 in India. This period includes varying economic conditions as well as changes in corporate governance practices that has impacted the institutional ownership conditions in India (Sarkar and Sarkar, 2000).

Our study contributes to blockholder governance literature in several important ways. First, we provide comprehensive categorisation of blockholders due to the ownership structure in India. While most studies in western economies focus on *pressure-resistant* and *pressure-sensitive* blocks (Park et al., 2008; Edmans and Holderness, 2016), we contribute to existing literature on blockholder heterogeneity by extending the classification to include *insider blockholders* as a distinctive group. Despite potentially large differences between blockholder types, relatively few studies account for blockholder heterogeneity in India. Second, we investigate whether family firm dominance, as manifested in ownership and control structures, affects governance role by diverse blockholder categories. Third, we contribute to the corporate governance literature by testing the Khanna and Palepu (2000) hypothesis on whether the 2009 regulatory intervention on improving transparency and information of insider ownership, impacts effective monitoring by blockholders. Such an investigation has direct relevance towards policy implications to “upgrade and harmonise” disclosure standards to meet global governance practices. Fourth, we contribute to the effect of blockholders in financial markets by providing empirical insights on short term market reaction (cumulative abnormal returns) towards block trades. While, most prior studies examine block purchases and sales independently (Barclay and Holderness, 1991; Park et al., 2008), we study market reaction to within-group (ex: pressure-resistant sells to another pressure-resistant) and across-group (ex: pressure-resistant sells to insider or pressure-sensitive category) block trades. Such combinations of trades has important implications towards interpreting incremental benefits of monitoring by particular blockholder categories. Finally, we extend previous findings that potential gains from added monitoring are greater when internal or external monitors are weaker (Park et al., 2008).

Our main findings are as follows: The three-day CAR for pressure-resistant and insider block trades is positive and significant (4.81% and 4.31% respectively), compared to pressure-sensitive blocks (-1.15%). Increase in pressure-resistant and insider block trades, has positive and significant impact on CAR than compared to increase in pressure sensitive block trades. Consistent with Park et al., (2008), trades by pressure-resistant blockholders are considered value enhancing, thus supporting the monitoring hypothesis, while trades by pressure-resistant blockholders are value reducing, implying the conflict of interest hypothesis (Pound, 1958). Insider blocks signal promoter’s confidence and incentive to monitor management (Sarkar and Sarkar, 2000). Our results indicate that the wealth effects surrounding family-owned firms is economically and significantly higher than family-owned-managed firms and non-family firms. Infact, pressure-resistant and insider block trades in family owned firms has significant impact on CAR than non-family firms. These results suggest an alignment of interest between pressure-resistant and insider categories in family owned firms explained by the large wealth at stake, and incentive to monitor (Shleifer and Vishny, 1986; Sarkar and Sarkar, 2000). However, consistent with Villalonga and Amit (2006), this result does not hold true for family owned and managed firms, where the promoter is either a CEO, MD or Chairman. Largely,

these results signify the role of pressure-resistant blockholders in professionally managed firms.

The CAR for post-2009 regulatory amendment is slightly higher than pre-2009 period. The multivariate analysis clearly show positive and significant impact of the 2009 regulatory amendment, pertaining to disclosures of insider shareholdings, on abnormal returns. Moreover, blocktrades by pressure-resistant and insider blockholders has positive and significant impact on CAR post 2009, thus supporting the contention that improved transparency and information impacts effective monitoring. Consistent with Park et al. (2008), results suggest that market perceives monitoring benefits by pressure-resistant blockholders as value-enhancing (value-decreasing), with lower (higher) percentage of independent (executive) directors on board. As board size increases beyond 8 directors, block trades by pressure-resistant category is considered value-enhancing. Moreover, increase in pressure-resistant category has significant market impact, in the absence of large shareholding ((blockholding > 5%). However, trades by insiders is considered value enhancing irrespective of large shareholding. Interestingly, leverage seems to have positive impact on abnormal returns in the absence of large shareholders as well as for trades by new blockholders.

The rest of the paper is organized as follows. In section 2, we provide a brief review of the corporate governance reforms in India. Section 3 discusses prior literature on blockholder heterogeneity and their role in governance, leading to various hypothesis development. Section 4 describes the data and methodology. In section 5, we discuss our results and section 6 concludes.

2. Ownership Disclosures and Corporate Governance Reforms in India

The institutional specificities of developing markets such as India are absence of well-developed capital and managerial market, less active takeover market, business practices that give greater importance to implicit trust-based contracting, and ownership structures that characterize business groups and family firms (Sarkar and Sarkar, 2012). The Indian corporate sector witnessed series of corporate governance reforms, that gathered momentum from late nineties. More specifically, the ownership disclosure requirements under Clause 35 underwent major reforms in March 2001, June 2006 and February 2009. The first round of reforms in 2001 differentiated the ownership structure in Indian firms into two major constituents – insiders (promoters) and outsiders (non-promoters) for all owners holding at least 1% of outstanding equity. An amendment to Clause 35 of the Listing Agreement in 2001 required disclosure of all shareholders holding more than 1% equity. Post April 2006⁵ the disclosure standards changed towards greater transparency, requiring complete disclosure of the identity and shareholding of all entities under Promoter and Promoter Group irrespective of any cut-off level, as well as greater detail of public shareholding under which institutions and non-institutions got reported separately. From February 2009⁶, the Equity Listing Agreement underwent two major amendments to Clause 35 and 41, regarding inclusion of details of shares

⁵ SEBI vide its circular no. SEBI/CFD/DIL/LA/2006/13/4 dated April 13, 2006 issued amendment to Clause 35 of Equity Listing Agreement.

⁶ SEBI vide its circular no SEBI/CFD/DIL/LA/2009/3/2 dated February 3, 2009 issued amendment to Clause 35 and Clause 41 of Equity Listing Agreement.

pledged by promoters and promoter group entities while disclosing shareholding pattern to stock exchanges and reporting the same in quarterly financial results of the company. In addition, Clause 35 was further amended to include each class of security as well as disclosures of total voting rights in the company⁷. More importantly, SEBI (Issue of Capital and Disclosure Requirements) Regulations, 2009⁸ further refined the term *promoter and promoter group* because the definition of promoters is “extremely critical for actions, regulations, research and analysis” (SMAC, 2004; Sarkar and Sarkar, 2012).

In this paper, we focus on the impact of the 2009 amendment aimed to enhance disclosures regarding ownership pattern, specifically pertaining to insider blockholders. Sarkar et al. (2013) commend that the set of regulatory amendments towards ownership disclosure requirements under Clause 35 during the period of 2000 enabled greater clarity in the definition of different ownership groups, and created more transparency about the identity of the ultimate owners of a listed company. This is very important and unique in the Indian context as compared to disclosure practices in other countries, because one could trace insiders having control rights in excess of cash flow rights⁹ due to direct and indirect equity stakes and pyramidal ownership structures. Such an exogenous regulatory intervention to improve disclosure norms and transparency, provides a unique opportunity to test the impact of important legislative changes on institutional ownership conditions as well as monitoring role by diverse blockholders in India.

3. Literature and Hypothesis Development

Our review of extant literature and hypothesis development is based on three main focus questions. First, why it is relevant to study market response to block trades, rather than firm value or performance. Second, why is it interesting to study diverse blockholders in Indian context? We justify this question in the light of *insiders* as an additional typology of blockholders, coupled with regulatory reforms and ownership structure prevalent in the Indian context that impacts costs and benefits of different blockholder categories. Third, we hypothesize that monitoring by outsider blockholders and internal governance mechanisms, are substitutes (not compliments) for one another. This argument holds true for existing external monitors (large pre-institutional holding) in the firm as well.

3.1 Market response to block trades by diverse blockholder categories

There is evidence that formations of blocks are associated with abnormal stock price increases (Mikkelsen and Ruback, 1985; Holderness and Sheehan, 1988; Barclay and Holderness, 1991, 1992). If blockholders do not impact cash flows that eventually accrue to

⁷ SEBI vide its circular no SEBI/CFD/DIL/LA/1/2009/24/04 dated April 24, 2009 issued amendment to Clause 35 of Equity Listing Agreement.

⁸ Regulations 2(1)(za) and (zb) of SEBI (Issue of Capital and Disclosure Requirements) Regulations, 2009 redefined and encompassed the definition of promoter and promoter group. For instance, promoter group included immediate relative of the promoter (i.e., any spouse of that person, or any parent, brother, sister or child of the person or of the spouse).

⁹ In Indian firms, insider control can be achieved with cash flow rights as less as 20%. The term control is defined under regulation 2(1) (c) of the SEBI Act, and includes the right to appoint majority of directors or to control the management or policy decisions.

minority shareholders, such stock price changes would be hard to explain. However, there are mixed results with firm value. Lins (2003) and Claessens et al. (2002) study emerging economies, and find that firm value is positively related to control rights held by outside blockholders. McConnell and Servaes (1990) and Mehran (1995), document no correlation between outside block ownership and firm value. These results suggest that institutional presence and firm value could be endogenous, because institutions are good at picking and investing in better-managed firms, and that firms with higher value attract blockholders. Choi (1991) suggests market reacts differently to trades by diverse class of blockholders, suggesting diverse block deal intentions, such as, monitoring effect, investment for private gain, potential business interests, takeover anticipation or undervaluation signaling. Therefore, we study short term market reaction (CAR) to block trade announcements in India, using diverse blockholder categories.

3.2 Blockholder Heterogeneity

Numerous studies focus on the impact of ownership concentration of large shareholders on firm performance, however assume ‘large’ shareholders to be indistinguishable among each other (Shleifer and Vishny, 1997; Makhija and Spiro, 2000; Morck, Wolfenzon, and Yeung, 2004). Edman (2013) argues that diverse blockholders impact firm performance in different ways due to varying incentives (Lehmann, Warning, and Weigand, 2004; Pedersen and Thomsen, 2003). Pound (1958) proposes competing hypothesis while considering incentives confronted by diverse blockholders: (i) the efficient-monitoring hypothesis predicts positive relation between institutional ownership and firm value because institutional investors have greater expertise and can monitor management at lower cost than small atomistic shareholders (ii) conflict-of-interest hypothesis predicts a negative relation between institutional ownership and the firm value as institutional investors are coerced into voting their shares with management because of other profitable business relationships with the firm (iii) the strategic-alignment hypothesis suggests that large shareholders and managers find it mutually advantageous to cooperate.

A growing body of corporate governance literature distinguish broadly between *pressure-resistant* investors, who have more economic incentive to monitor, vs *pressure-sensitive* investors, who have less economic incentive to monitor¹⁰ (Brickley, Lease, and Smith, 1988; Almazan, Hartzell, and Starks, 2005 and Agrawal, 2012). In India, the diversity of blockholders can be attributed to *insider vs outsider blocks* where insiders are promoters¹¹. In this relation, we follow, Brickley, Lease, and Smith (1988); Hutchinson, Seamer and Chapple (2015), and Muniandy, Tanewski, and Johl (2016), and extend the broad typology of blockholders into three categories, *pressure-resistant*, *pressure-sensitive* and *insider blocks*.

First, among *pressure-resistant* blocks we observe two categories; 1) investment firms, such as, hedge funds, mutual funds, pension funds, private equity, investment advisory and brokerage firms, foreign institutional investors, and financial institutions, such as commercial

¹⁰ This compelling distinction is driven by theory, not by actual observed evidence of monitoring because it relies on recognizing significant (potential) business relationships that investors have with block firms.

¹¹ Promoter is widely used in India to refer to any person or body corporate who directly or indirectly controls a firm. Regulations 2(1)(za) and (zb) of SEBI (Issue of Capital and Disclosure Requirements) Regulations, 2009 (hereinafter referred to as ‘ICDR Regulations’) define promoter and promoter group.

banks and insurance companies; 2) individual investors (includes HUF). These blockholders are in better position to actively engage in monitoring activities due to their fiduciary responsibility to protect their investments. Chen et al. (2007) show that independent financial institutions with long term investment perspective have a stronger incentive to monitor with arguably low cost of monitoring. In a related work, Park, Selvili, and Song (2008) find significant abnormal returns for activist and strategic blocks suggesting the existence of monitoring benefits. However, there are varied arguments with regards to commercial banks and insurance companies, as these institutions might have potential business interest in block firms (Brickley et al., 1988). There is also counter argument and evidence that banks¹² have relative benefit in monitoring corporations due to their ability to access inside information (Lehmann and Weigand, 2000).

Next, we focus on *corporations* as part of *pressure-sensitive* blocks. Corporations could be private or public corporations, within same industry or business groups. Pressure-sensitive blockholders are largely those who might have current or potential business relations with the firm. Corporate blockholders enjoy certain private benefits not available to institutional and individual blockholders. For example, corporate blockholders enjoy synergistic benefits in terms of market collaborations between the purchasing firm and the invested firm (Allen and Phillips, 2000; Fee et al., 2006; Barclay et al., 2009). These blockholders might not be as effective in monitoring and play a less important governance role since they would be less likely to object questionable practices for fear of losing business (Ferreira and Matos, 2008; Chen et al., 2007). Also, corporate blockholders are different from other types of blockholders since they also suffer from agency problems and require effective monitoring mechanisms themselves. Barclay and Holderness (1991) argue that corporate blockholders rely on their representatives on the boards of their invested firms to pursue their private benefits of control. Therefore, we classify corporations as pressure-sensitive blocks.

Finally, we categorise the *insider* blockholders (promoters, promoter group companies, promoter trusts, and insider directors) as a separate group, since they are neither independent nor pressure-sensitive investors. The traditional outlook towards insiders associated with the firm may use ownership as an entrenchment device to extract private benefits. However, given a large amount of wealth at stake, these large insider shareholders have the incentive to monitor actively, scrutinize management, and curb rent extraction (Shleifer and Vishny, 1986; Sarkar and Sarkar, 2000). Moreover, promoters with large amounts of assets in the form of securities and their actions could also serve as a strong signalling mechanism to minority investors.

The above discussion and typology of blockholders leads us to highlight our main hypothesis as follows.

H1a: Block trades by *pressure-resistant* blockholders will be positively associated with abnormal returns (CAR).

¹² In our study we include banks and insurance companies among investment firms as part of pressure-resistant blocks. The argument of banks having potential business interests would hold good for commercial banks, which brings the focus on debt overhang issues, and in such cases the bank is generally not interested in monitoring, rather focus on recovering the debt. However, our sample of banks consists mainly of foreign banks and fewer local banks and insurance companies, hence we find it logical to add banks and insurance companies to pressure-resistant category of blockholders.

H1b: Block trades by *pressure-sensitive* blockholders will be negatively associated with abnormal returns (CAR).

H1c: Block trades by *insider* blockholders will be positively associated with abnormal returns (CAR).

3.3 Regulatory reforms

Khanna and Palepu (2000) argue that monitoring by large shareholders in developing countries may not be as effective as in developed countries because of the poor availability of information due to inadequate disclosure norms and weak enforcement. The set of reforms discussed in section 2, especially towards ownership disclosure requirements under Clause 35 and 41 during 2009, is mainly to enable transparency and access to information regarding identity of different ownership groups on public domain through quarterly financial reporting. This exogenous regulatory amendment in 2009 provides a natural setting to test its impact on the monitoring role of blockholders, as perceived by market reaction. Hence, we conjecture that the regulatory amendments in 2009 would enable effective monitoring by pressure-resistant and insider blockholders simply due to availability and transparency of information in public domain post 2009.

H2: Block trades by *pressure-resistant* and *insider* blockholders will be positively associated with abnormal returns post-2009 compared to pre-2009 due to regulatory amendments.

3.4 Family owned and managed firms

The objective of examining the impact of diverse blockholders on CAR, with particular focus on family owned and managed firms, is because these firms have ownership and governance structures that are distinctly different from those of widely held firms (Anderson and Reeb, 2004). Whether family firms are more or less valuable than non-family firms, remains an open question¹³. The typical portrayal suggests that these organizations suffer from inter-generational squabbles, executive entrenchment, and nepotism, all of which may detract from performance¹⁴. This traditional outlook towards family firms supports the entrenchment hypothesis (Morck et al., 1988) implying opportunistic behavior by founding family members as manifested in expropriating firms' resources through excessive compensation, special dividends and perquisites, and self-dealings (Anderson and Reeb, 2003; Masulis et al., 2011). Luo and Chung (2012), and Sarkar and Selarka (2015) suggest that the ability to expropriate are particularly strong when families hold key positions on the board, namely as a CEO, or chairperson or both.

Pagano and Röell (1998) note the presence of other large blockholders can reduce the concerns of controlling shareholder wealth expropriation. Maury and Pajuste (2005) find that

¹³ In studies of the S&P 500, Business Week 1000, and Fortune 500 samples, respectively, Anderson and Reeb (2003, 2004), McConaughy et al. (1998) and Villalonga and Amit (2006b) report that the Tobin's q of family firms is greater than that of other corporations. There is contrasting evidence from Europe and Asia (Claessens et al., 2002; Cronqvist and Nilsson, 2003; Maury, 2006). In India, some studies suggest family firms perform better than nonfamily firms (Jackling and Johl, 2009; Sarkar and Selarka, 2015).

¹⁴ For detailed literature refer, Allen and Panian, 1982; Chandler, 1990; Gomez-Mejia et al., 2003; Perez-Gonzalez, 2006; Schulze et al., 2001, 2003.

family firm performance is enhanced if the second blockholder is a large outside shareholder. However, Villalonga and Amit (2006) find that outside blockholders fail to be effective monitors over family managed firms. In contrast, the alignment of interest hypothesis postulates that higher family ownership results in greater alignment of interest between the controlling owners and the other stakeholders (Demsetz and Lehn, 1985), resulting in lower expropriation by family owners (Ali et al., 2007). That is, given a large amount of wealth at stake, these insider shareholders themselves have the incentive to monitor actively, scrutinize management, and curb rent extraction (Shleifer and Vishny, 1986; Sarkar and Sarkar, 2000).

The above review of literature brings to focus that the role of blockholders' governance may vary in family owned vs family owned and managed firms. Therefore, we expect that market reacts positively to block trades by pressure-resistant and insider blockholders in family owned (and not managed) firms compared to non-family firms, assuming there is alignment of interest between pressure-resistant and insider blockholders in monitoring management. However, the role of pressure-resistant blockholders is reinforced in family owned and managed firms, in comparison to insider blockholders, if one considers that entrenchment is more when family has management control as well. The literature reports evidence of varying threshold being used to classify family owned firms (La Porta et al., 1999; Claessens et al., 2000). Following Anderson and Reeb (2003), we define family owned firms if the promoter control in terms of voting rights is atleast 20 percent. We include a criterion for family owned and managed firms, where the promoter ownership is > 20%, and the promoter is either a chairman, managing director, or chief operating officer in the company.

H3a: Block trades by *pressure-resistant* and *insider* blockholders in family owned firms will be positively associated with abnormal returns, compared to non-family firms.

H3b: Block trades by *pressure-resistant (insider)* blockholders in family owned and managed firms will be positively (negatively) associated with abnormal returns, compared to non-family firms.

3.5 Internal Governance mechanism

The market's reaction to block trades would depend on existing governance mechanisms. Park et al. (2008) hypothesise that monitoring by outsider blockholders and internal governance mechanisms are substitutes (not compliments) for one another, and thus, the potential gains from added monitoring are greater when internal monitors are weaker. There are fewer agreements regarding the optimal governance mechanisms such as board size or board independence (Coles et al., 2008; Adams and Ferreira, 2007). Sarkar and Sarkar (2000) and Bennedsen, Kongsted, and Nielson (2008) acknowledge a piece-wise relation between optimal board size and firm value, where optimal board size may be linked with various firm characteristics such as size, age, and industry affiliation as well as unobserved factors. Ghosh (2006) find evidence that larger boards lead to poorer performance (market-based as well as in accounting terms) in India. In regard to board independence, Sarkar and Sarkar (2009) document that busy independent directors add value to a firm by helping the management with their expertise and external social connections. Haldea (2010) suggests that independent directors in majority of Indian companies are appointed by the management and the promoters largely to fulfil regulatory norms rather than to be active monitors.

Existing literature highlights that one of the channels of expropriation of controlling shareholders is through executive positions on the board (Fan and Wong, 2002; Bertrand et al., 2003; Friedman et al., 2003). In India there is prevalence of insider dominated board of directors, such as promoter and executive directors on board, who enjoy substantial ownership and control and often hold top executive positions with the objective of controlling the firm (Carney and Gadajlovic, 2002; Gomes-Mejia et al., 2003). Given this background, we consider board size, percentage of independent and executive directors as proportion of total directors on board, as important internal monitoring mechanism.

H4: We hypothesize negative association between abnormal returns and pressure-resistant blocks when interacted with higher proportion of independent directors, lower proportion of executive directors, and optimal board size.

3.6 Institutional holding

Another important external monitoring mechanism is the presence of large institutional investors. There have been several papers taking the view that large institutions mitigate agency problems by behaving as monitors (e.g., Admati, Pfleiderer, and Zechner, 1994; Huddart, 1993; Maug, 1998; Shleifer and Vishny, 1986). There has also been considerable empirical support for this theory (e.g., Agrawal and Mandelker, 1990; Bertrand and Mullainathan, 2001; Chung, Firth, and Kim, 2002; Hartzell and Starks, 2003). From an agency perspective, the presence of institutional investors is generally regarded as beneficial. However, if the block firm already has one or more large blockholders prior to the block trade, then the incremental monitoring-related benefit from an additional blockholder is likely to diminish. Shleifer and Vishny (1986) and Maug (1998) argue in favour of lower incremental benefits from additional as well as new blockholder in the presence of large incumbent blockholders. Thus, we include two proxy variables to test for the effect of prior institutional holding on market reaction to blockholders. First, we control for large institutional holding during the quarter prior to the block trade, where large is defined as more than 5% equity ownership. Second, we differentiate between old and new (first time) blockholders, where new blockholder is defined as those investors who did not have any prior shareholding in block firms for past one year.

H5: We hypothesize negative association between abnormal returns and pressure-resistant blocks in the presence of large institutional holding.

3.7 Block Characteristics and Financials

Consistent with prior literature we examine the market reaction to block trades using various controls. First, we expect block size (in percentage terms with respect to outstanding shares) to be a significant explanatory factor for the wealth effect. Greater the proportion of block ownership, larger is the likelihood that the benefit of monitoring will exceed the cost (Park et. al., 2008). Therefore, we hypothesize a positive relationship between block size and market reaction to pressure-resistant and insider blocks. Second, we control for block's financials such as, market capitalisation, price to book ratio and leverage. The literature suggests decrease in firm value as firms becomes larger and more diversified (Lang and Stulz,

1994; Sarkar and Sarkar, 2000), and leverage¹⁵, which is the proportion of total debts to total assets, could lead to external corporate control (Hutchinson and Gul, 2004; Chen and Jaggi, 2000). Therefore, we hypothesize that for small sized firms, pressure-resistant blocks would have a greater positive effect on market reaction as compared to pressure sensitive blocks. If financial leverage and block trades by pressure-resistant blockholders are viewed as substitute monitoring, we expect that as leverage increases, pressure-resistant blockholders would have lesser positive effect on short term market returns. Moreover, if price to book ratio could be interpreted as undervaluation signalling effect (Choi, 1991), we expect block trades by pressure-resistant blockholders and insiders on low PB firms to have positive impact on the market. We control for the firm's industry as institutional investors' preference for investments may be driven by preferences for particular industries (Hartzell and Starks, 2003), and year effects from 2005 – 2015.

4. Data, Variables and Methods

4.1 Data

Our sample consists of block trades (buy and sell) for ten-year period from December 2005—March 2015, comprising of 636 Indian listed firms, that excludes financial companies. SEBI vide its circular no. MRD/DoP/SE/Cir-19/05 dated September 2, 2005 issued guidelines for execution of block trades on the stock exchanges, therefore our data include trades post December 2005. The data on block trades is obtained from CMIE Prowess that includes information on the firm name, deal date, client name, block size, price per share, and value of block. We further obtain detailed institutional ownership, financial data and stock price data from the Prowess database.

Our initial sample consists of 2547 buy-side and 2320 sell-side block trades, that accounts for multiple block trades on a given date, by all possible blockholder categories during 2005—2015. Since our primary variable of interest is cumulative abnormal returns (CAR) around the date of block trade, our sample for CAR calculation is finally 1065 unique transactions due to the following reasons: There were multiple entries corresponding to the same block deal on the same date by the same blockholder, these multiple entries were removed to obtain a unique deal and date combination for the event study. Secondly, some data had missing closing prices and stock returns were not available for the estimation window, $t-150$ to $t+10$. Finally, Sensex returns were not available for the defined estimation window.

However, in case of the regression analysis, where we account for multiple trades by different blockholder categories (on the same firm), this sample increases to 1447 buy-side and 1725 sell-side block trades. This number is based on stock prices for the period $t-150$ to $t+10$. Further, the sample for various model specifications reduces to around 1320 block trades that has complete information on board of directors, financial variables, and ownership structure. In case of sub-sample analysis, our data further reduces to 1213 block trades. We also obtain incorporation year, business group affiliation and national industrial classification (NIC) from CMIE-Prowess. Our dependent and independent variables are described in the section below.

¹⁵ Related literature emphasizing the role played by debt in reducing agency conflicts between managers and shareholders are Dewatripont and Tirole (1994), Jensen and Meckling (1976), Grossman and Hart (1982), Stulz (1990), and Hart and Moore (1995).

4.2 Variables

The variables used in our empirical analysis can be grouped into three categories, (i) dependent variables measuring market reaction to block trades (3- day cumulative abnormal returns, CAR) (ii) primary variable of interest that classifies blockholders into pressure-resistant, pressure-sensitive and insider blocks (iii) secondary variables of interest that includes regulatory and ownership structure of block firms (iii) other control variables such deal size, corporate governance mechanisms, prior institutional holding and firm-level financials which might also affect market reaction to block trades.

4.2.1 Dependent Variable – Cumulative Abnormal Returns (CAR)

We consider cumulative abnormal returns (CAR) around block trade by diverse blockholder categories as our event of interest because it leads to the creation of a new monitoring mechanism and its economic effect can be measured by the event study methodology. We calculate cumulative abnormal returns (CAR) using a 3-day event window defined as -1, 0, and +1 days relative to the date of block deal (purchase and sale), using estimation window as (-150, -30) days. Announcements that had missing returns on any of the three days were dropped from the analysis. We also vary the event window to (-2, 2), (-3, 3), (-5, 5) and (-10, 10) days, and estimation window to (-180, -30) and (-200, -30) days, to check for robustness of results (not reported). We use Fama-French 3-factor model (Fama and French, 1993) to calculate CAR. We consider the risk-free rate to be 7.38% and the 3 factors are drawn from Varma et al. (2013), adjusted for survivorship bias. The model is given as follows:

$$R_{jt} - R_f = \alpha_j + \beta_1(R_{mt} - R_f) + \beta_2SMB + \beta_3HML + \varepsilon_{jt} \quad (1)$$

where R_{jt} is the return for stock j in period t , R_f is the risk-free rate of return, $(R_{mt} - R_f)$ is the market premium, SMB measures the historic excess returns of small market capitalization stock over big market capitalization stocks, HML is the historic excess returns of value stocks over growth stocks.

4.2.2 Independent Variable - Blockholder Categories

Our main independent variable is blockholder categories. Prowess provides a classification of shareholding data into promoter and non-promoter names for block deal firms. We use this data to categorise blockholders into four main categories: 1) individual (includes HUF), 2) investment (mutual funds, investment advisory and brokerage firms, pension funds, hedge funds, private equity, foreign Institutional Investors, commercial banks and insurance companies, 3) corporations (private and public companies, within same industry or business groups), and 4) insiders (includes promoters, promoter companies, promoter trusts and executive/promoter directors). This classification is further re-grouped into pressure-resistant, pressure-sensitive and insider blockholders, where pressure resistant category includes individuals and investment firms, and pressure sensitive category comprises of corporates.

Following prior studies on blockholder voice and exit (Edman and Manso, 2008; Bharath, Jayaraman, and Nagar, 2013; Edmans, 2014), we estimate CAR around block purchases and sales for pressure-resistant, pressure-sensitive and insider blockholder categories. However, these results do not account for trades across (and within) block categories which might have

implications for the monitoring hypothesis. To elaborate, we expect market reaction to be different if a pressure-resistant blockholder buys from another pressure-resistant vs if the purchase is from pressure-sensitive or insider category. To account for such combination of trades, we aggregate buy (and sell) trades across blockholder categories, and define a measure called “delta trades” that accounts for increase in block purchases by pressure-resistant, pressure-sensitive and insider block trades, across blockholder categories. For example, the measure of pressure-resistant delta trade is all trades for which the buyer was pressure-resistant blockholder and seller was either pressure-sensitive or insider blockholder. This detail excludes trades where the buyer and seller belong to the same blockholder category. Therefore, delta trades account for increase in block purchases by one category with respect to block sales by other categories. The number of block trades for each blockholder category is given in Table 3, Panel B. The largest buy-group is insiders, (545 purchases), followed by pressure resistant (471 buys) and pressure sensitive category (313 buys). The largest sell group is pressure-resistant (617 sales), followed by insider (478 sales) and pressure-sensitive (234 sales). The delta trades inform pressure-sensitive reports highest across-group trades, followed by insider and pressure-resistant.

4.2.3 Regulatory Intervention

We use 2009 amendments to Clause 35 and Clause 41 of the Equity Listing Agreement regarding disclosures of shareholding of promoter and promoter group entities, and reporting the same in quarterly financial results of the company, as a regulatory control. The variable is defined as Regulation2009, which is a dummy taking a value 1, when the year is greater or equal to 2009 and 0 otherwise.

4.2.4 Ownership Characteristics

We classify firms into family-owned only, family owned and managed vs non-family firm. Following Anderson and Reeb, (2003), we define family-owned as those firms where the promoter control (in terms of voting rights) is greater than (or equal to) 20 per cent. Following, Sarkar and Selarka (2015), we define family owned and managed firms as promoter holding more than 20 percent of the shares and the promoter is either the chairman, managing director, or chief operating officer in the firm.

4.2.5 Corporate Governance

The Prowess database contains information on directors from company annual reports and corporate governance reports, that includes names of directors, director’s designation as executive, promoter or independent¹⁶, as well as the identification of whether the promoter of the company holds management positions of CEO or chairperson on the board. With this information, we find various corporate governance characteristics of the firm, namely, board size, percentage of promoter directors, executive directors, and independent directors on board. For our analysis, we use a piece-wise linear specification of board size which comprises of two

¹⁶ Directors are classified as independent if they do not hold executive position, or have not held executive position for last three years, do not hold one per cent or more ownership and are not related to the promoter/s of the firm.

components. The first component is defined as total number of directors on board. The second component is a difference between the board size of the firm minus the median board size, interacted with a dummy variable that takes the value 1 if the board size of the firm is greater than the median board size and 0 otherwise. In our sample, the median board size is 8 directors.

4.2.6 Prior Institutional Holding

We also collect information on institutional holding during the quarter prior to the block trade to access pre-institutional holding in block deal firms. Prowess provides data on equity holding divided into promoter and non-promoter group. We consider institutional holding from non-promoter group that includes, mutual funds, banks, development financial institutions, insurance companies, foreign institutional investors, venture capitalists, corporate bodies and other non-promoter institutions. We define large institutional holding as institutional holding of greater than 5% during the quarter prior to the block trade. Following Edmans and Holderness (2016), we also explore the effect of institutional holding of greater than 1% and less than 5%, since SEBI regulation on amendment to Clause 35 of the Listing Agreement in 2001 required disclosure of the identity of all shareholders holding more than 1%. Next, we distinguish between new or first-time blockholders versus old blockholders (those who already hold stake in the company). We define new blockholders as those who do not have prior holding in the firm for past one year before the block deal.

4.2.7 Firm level Financials

The above sample of data is then matched with the information from Prowess on financials, which include market capitalization, Price to Book (PB) ratio, and Leverage (measured by total debt as a proportion of total assets) for the quarter before the block deal.

The definition of the variables is given in Table 1.

4.3 Multivariate Analysis

Our dependent variable is 3-day cumulative abnormal return (CAR), corresponding to the Fama-French 3-factor model. Our primary independent variable is blockholder categories described as, pressure resistant, pressure sensitive and insider blocks. In our benchmark model as shown in equation 2, we control for block deal characteristics (percentage deal size), corporate governance characteristics (board size, percent of independent, executive and promoter directors on board) and firm level financials (market capitalisation, PB and leverage). We vary the benchmark model by considering block sales and delta trades for every blockholder category.

$$CAR = a + bPressure\ Resistant_{buy/sell/delta} + cInsider_{buy/sell/delta} + dPercentage\ Deal\ Size + eBoardsize + fBoardsize_{piece} + gIndependent\ Directors + hExecutive\ Directors + iMarket\ Cap + jPB + kLeverage + Industry\ Dummies + Year\ Dummies + \varepsilon \quad (2)$$

For the rest of the analysis, blockholder delta is our main independent variable that captures across-group increase in block purchases. We add the effect of regulation 2009, ownership structure (family-owned and family owned-managed firms) and institutional

holding (large pre-institutional holding and new blockholders) to later models, alongside other controls. We present the comprehensive model specification in equation 3, while we add or remove variables to test various hypothesis described in section 3.

$$CAR = a + bPressure\ Resistant_{delta} + cInsider_{delta} + dPercentage\ Deal\ Size + eBoardsize + fBoardsize_{piece} + gIndependent\ Directors + hExecutive\ Directors + iMarket\ Cap + jPB + kLeverage + lRegulation2009 + mFamily_{owned} + nFamily_{ownedManaged} + oLarge\ Institutionalholding + pNew\ Blockholder + qIndependent\ Director * Pressure\ Resistant_{delta} + rIndependent\ Director * Insider_{delta} + sExecutive\ Director * PressureResistant_{delta} + tExecutive\ Director * Insider_{delta} + Industry\ Dummies + YearDummies + \varepsilon \quad (3)$$

We also perform various subsample analysis to test different hypothesis described in section 3. For example, we divide the sample for pre-2009 regulation vs post-2009 regulation and family owned (and managed) vs non-family owned (and managed firms) to substantiate our finding regarding hypothesis 2 and 3. We interact corporate governance characteristics and large shareholding with pressure resistant and insider categories of blockholders to test hypothesis 4 and 5. In this case too, we conduct subsample analysis for large shareholding vs < 5% shareholding , and new vs old blockholders. The results are discussed in Section 5.

5. Results

5.1 Announcement Effects

Table 2 reports descriptive statistics for all block firms with announcement of block trades being the unit of analysis. All variables (except dummy variables) are winsorized at 1% and 99% levels. Table 3 Panel A reports CAR for different event time periods varying the estimation window. The block trades in our sample are associated with average abnormal stock-price increase of 1.37% for the 3-day CAR. Abnormal returns reduce as our period increases to 5-day and 7-day CAR, and is negative at 11-days and 21-days with varying levels of significance. Table 3 Panel B presents CAR for buy, sell and delta trades across different blockholder categories; three-day CAR for buy trades for pressure-resistant (2.06%) and insider (2.3%) blockholders is significantly positive, while pressure-sensitive blockholders is negative (-0.5%) but not significant. The positive market reaction to pressure-resistant and insider blockholders suggest that they add value to the block, thus consistent with efficient monitoring hypothesis (Pound, 1958) and that promoter ownership may serve as a strong signalling mechanism (Sarkar and Sarkar, 2000). Contrary to the literature on blockholder exit, CAR for sell trades appear positive and significant for pressure-resistant (0.79%), pressure-sensitive (3.61%) and insider (1.5%) blockholders. This raises an important question about who the blockholder is selling to (or buying from), leading to observation about within and across-group block trades. Hence, we account for market reaction to delta trades that implies increase in across-group block purchases. The results suggest that market perceives increase in pressure-resistant (4.81%) and insider (4.31%) trades as value-enhancing, while increase in pressure-sensitive trades (-1.15%) show negative CAR.

To elaborate the results on delta trades, we report mean CAR for within-group and across-group block trades in Panel C of Table 3. The results highlight that mean CAR is positive and

significant when the pressure-resistant blockholder purchases blocks from either pressure-sensitive or insider category (across-groups), suggesting increased monitoring benefits from pressure-resistant blocks. Similarly, market reacts positively to increase in insider block purchases from pressure-resistant and pressure-sensitive categories. However, CAR is negative for block purchases by pressure-sensitive category, especially when the seller is pressure-resistant blockholders. Interestingly, the mean CAR does not reflect any significant announcement effect when the block deal occurs between same category of clients. These results substantiate the importance of across-group trades, hence going forward, we use CAR from delta trades as our main dependent variable.

In Panel D, we report CAR for various institutional and ownership characteristics of block firms. Market reaction to post-2009 regulatory amendment shows greater significant impact than pre-2009, however without much economic difference. The wealth effects surrounding family-owned firms (718), family owned and managed firms (457), and just family managed firms (476), is economically and significantly higher than non-family firms. The results show marginal difference in CAR for old vs new blockholders, and large blocks versus blocks with less than 5% shareholding. In Table 4, we report the correlation among different variables used in the study. While some of the independent variables show significant correlation, the magnitude of correlation is low.

5.2 Determinants of CAR

We relate blockholder categories (pressure-resistant, pressure-sensitive and insider) to 3-day CAR while taking into account regulatory amendment, ownership structure (family-owned and family-owned-managed firms) and prior institutional holding (large institutional holding and new blockholders). We control for block size, corporate governance characteristics (board size, percent of independent, executive and promoter directors on board), and firm level financials (market capitalisation, PB and leverage) of block firms.

Table 5 presents the baseline results as we test the effect of pressure-resistant and insider blockholder categories on CAR for buy, sell and delta trades, while considering pressure-sensitive as the control group, and controlling for block size, existing corporate governance and financial indicators. In column 1, the coefficient of pressure-resistant and insider blockholders, is positive and highly significant (at 1% levels) compared to pressure-sensitive corporates. This result is consistent with Chen et al. (2006) that independent institutions with long-term investments specialize in monitoring and influencing firms. Results support the contention that insiders' due to large stakes have incentive to monitor actively, scrutinize management, and curb rent extraction, and also signal minority shareholders (Shleifer and Vishny, 1986; Sarkar and Sarkar, 2000). Deal size (percentage) does not appear significant. Increase in board size has positive effect on CAR, however the coefficient turns negative when board size increases greater than 8 directors (median board size in our sample), thus consistent with "more is not necessarily better" (Yermack, 1996; Bhagat and Black, 2001). Market reaction to block purchase is negatively correlated with proportion of executive directors on board, supporting the concern of expropriation by controlling shareholders through executive positions on the board (Bertrand et al., 2003; Perry and Peyer, 2005). The effect of independent directors as outsider monitors on market reaction is mostly insignificant. This result supports Haldea (2010) that independent directors in a majority of Indian companies are appointed to

fulfil regulatory norms rather than to be active monitors. Among financial control variables, small sized firms (lower market capitalisation) has positive effect on CAR, while price to book ratio and leverage appears insignificant.

In column 2, we explore wealth effects for sell trades, and consistent with exit literature, the relation between pressure-resistant (and insider) sell trades and CAR is negative, although insignificant. In column 3, we account for delta that implies across-group increase in increase in pressure-resistant, pressure-sensitive and insider blockholders. Results indicate that increase of pressure-resistant and insider category of blockholders had significant and positive impact on CAR, compared to pressure-sensitive blockholders, thus consistent with blockholder governance literature on monitoring and conflict-of-interest hypothesis.

In Table 6, we introduce regulatory dummy that captures amendment in 2009 to the Equity Listing Agreement regarding disclosures of shareholding. In Column 1, the regulatory dummy is positive and significant (at 5% level), indicating positive impact of regulatory amendment on CAR post 2009. The effect of pressure-resistant and insider blockholders on CAR remains positive and significant. In Column 2 and 3, we show sub-sample analysis of block trades before and after 2009 respectively. We find the effect of pressure-resistant and insider blockholders on CAR is greater post 2009, than in the pre-regulatory period. These results support Khanna and Palepu (2000) that monitoring by blockholders can be effective with improvement in adequate disclosures and availability of information.

In Table 7, we examine the effect of ownership structures characterized by family owned firms (promoter holding is greater than 20 per cent), and family owned and managed firms (promoter holding is greater than 20 per cent, and the promoter is either the chairman, or CEO or MD) firms on block trades by diverse blockholder categories. In column 1, although family owned firms itself does not have significant impact on CAR, increases in block purchases by pressure-resistant and insider blockholders has positive and significant impact on CAR. This result is supported in the sub-sample analysis of family-owned (but not managed) firms (column 2) vs non-family owned firms (column 3). This result is consistent with alignment of interest hypothesis that postulates higher family ownerships results in greater alignment of interest between the controlling owners and the other stakeholders (Demsetz and Lehn, 1985). Moreover, given a large amount of wealth at stake, these large insider shareholders have the incentive to monitor actively, scrutinize management, and curb rent extraction (Shleifer and Vishny, 1986; Sarkar and Sarkar, 2000).

Next, in column 4 we introduce the dummy variable family owned and managed firms (where promoter owns atleast 20% equity and is either a chairman, CEO or MD). The results show that block trades by pressure-resistant and insiders blockholders as positive and significant, while controlling for family owned and managed firms, while these firms themselves do not have significant effect on CAR. Contrary to our hypothesis, the sub-sample analysis in column 5 and 6 show that market reaction to pressure-resistant and insider blockholders is more significant in non-family firms, than family owned and managed firms. This result supports the contention outside blockholders fail to be effective monitors over family managed firms (Villalonga and Amit, 2006). Moreover, there is not significant result associated with trades of insiders in the family owned and managed firms. Overall the results also suggest that pressure-resistant and insider blockholders produce positive wealth effects in professionally managed firms than family managed firms.

In Table 8, we test the hypothesis that monitoring by outsider blockholders and corporate governance mechanisms are substitutes (not compliments) for one another. In column 1, we interact pressure-resistant category of blockholders with board size piece, and percent of executive and independent directors. The co-efficient for interaction between pressure-resistant blockholders and independent directors is negative and highly significant, while positive and significant for interaction with executive directors. Thus, consistent with Park et al. (2008), the results imply that, as executive directors increase on board (or independent directors decrease on board), the market perceives monitoring benefits from pressure-resistant blockholders as value-enhancing, because the internal corporate governance mechanism is weak. Similarly, as board size increases beyond median threshold of eight directors, the incremental benefits from pressure-resistant blockholders is reducing. The sub samples for pressure-resistant (column 2) and insider (column 3) blockholders support the result that the effect of independent directors on abnormal returns lowers in the presence of pressure-resistant and insider blockholder groups.

In Table 9, we test the presence of external monitoring through prior institutional holding while evaluating the market reaction to blockholders. In column 1, we first control for large institutional holding (greater than 5% ownership) during the quarter prior to the block trade. While large institutional holding itself does not have impact on CAR (column 1), we find that trades by pressure-resistant and insider blockholders are perceived as value enhancing in firms without large institutional holding (column 3). Whereas we find no impact of pressure-resistant and insider block trades on CAR in firms with large shareholding (column 2). These results support Shleifer and Vishny (1986) and Maug (1998) that incremental monitoring-related benefit from an additional blockholder in the presence of large shareholding is likely to diminish. In column 4, we examine the effect of new (first time) blockholders who do not have any holding in the firm for past one year prior to the block trade. While, new blockholders themselves do not impact CAR, however block trades by pressure-resistant new blockholders has a positive and significant impact on CAR (column 5), thus indicating incremental positive monitoring benefits due to formation of new blocks.

6. Conclusion

In this paper, we examine market reaction to block trades by different blockholder categories in a developing market setting characterized by dominance of family-owned and managed firms, and regulatory amendments regarding ownership disclosures. Our event study results suggest the three-day CAR corresponding to pressure-resistant blocks is positive and significant than pressure-sensitive corporates. Market reacts positively to increase in insider block purchases. Interestingly, the market does not react significantly in case of buy and sell between same category of blockholders. The wealth effects surrounding family-owned firms is economically and significantly higher than family-owned-managed firms and non-family firms. Results further suggest positive and higher CAR post regulatory amendment than pre-period.

Next, we test the impact of blockholder categories (pressure-resistant, pressure-sensitive and insider categories) on 3-day CAR while controlling for regulatory amendment, family owned and managed firms, block's corporate governance characteristics, prior institutional

holding, deal size (percentage) and block's financials. Consistent with univariate results, increase in pressure-resistant and insider block trades, have larger positive and significant market reaction, however increase in pressure-sensitive blocks shows negative impact on market. These findings could be attributed to monitoring benefits of large independent investment blocks compelled with investment motives by individual blocks, compared to conflict-of-interest hypothesis for pressure-sensitive blockholders. Given a large amount of wealth at stake, insider blocks have the incentive to monitor actively, scrutinize management, and curb rent extraction, thus signalling the market conviction about their own firms.

The novel contribution of the paper to blockholder governance literature takes into account ownership structures and regulatory interventions unique to the Indian set-up. Results indicate the positive impact of SEBI's corporate governance regulatory reforms in 2009 regarding disclosures of ownership holding, specifically pertaining to promoter and promoter group. Moreover, block trades by pressure-resistant and insider blockholders has positive and significant impact on CAR post 2009. While family firms itself does not show significant impact on abnormal returns in the multivariate analysis, we find block trades by pressure-resistant and insider category of blockholders in family-owned and professionally managed firms (promoter holding > 20%) has positive and significant impact on CAR. However, this result does not hold true for family-owned and managed firms where the promoter is either a CEO, MD or Chairman.

Consistent with Park et al. (2008), results suggest that market perceives monitoring benefits by pressure-resistant blockholders as value-enhancing (value-decreasing), with lower (higher) percentage of independent (executive) directors on board. As board size increases beyond 8 directors, block trades by pressure-resistant category is considered value-enhancing. While large shareholding (blockholding > 5%) itself does not have an impact of abnormal returns, the market reacts positively to increase in trades by pressure-resistant category in the absence of large shareholders. These results are consistent with Shleifer and Vishny (1986) and Maug (1998) on the incremental contribution of new pressure resistant blockholders to actively engage in monitoring activities. However, trades by insiders is considered value enhancing irrespective of presence or absence of large shareholding. Interestingly, leverage seems to have positive impact on abnormal returns in the absence of large shareholders as well as for trades by new blockholders.

Among corporate governance controls, increase in board size (upto 8 directors) has positive and significant impact on CAR, followed by reduced market impact. A larger representation of executive directors on board lowers abnormal stock returns, highlighting concerns of expropriation by managers on board. We do not find any significant effect of independent directors on abnormal returns, thus consistent with the argument that independent directors on Indian boards are largely present to fulfil regulatory norms., Small sized firms have greater positive effect on market reaction. However, deal size, leverage and price to book ratio itself do not appear significant in the analysis.

Future research could be directed towards validating impact of monitoring by pressure-resistant and insider blockholders on firm value, by relating accounting as well as market-based measures of performance to diverse blockholder categories.

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Table 1: Variable Definition

Variable Name	Definition	Source
<i>Dependent Variables</i>		
CAR	Cumulative Abnormal Return where the abnormal return is calculated using Fama-French 3 factor model: $R_{jt} - R_f = \alpha_j + \beta_1(R_{mt} - R_f) + \beta_2SMB + \beta_3HML + \varepsilon_{jt}$ where R_{jt} is the return for stock j in period t, R_f is the risk free rate of return, $(R_{mt} - R_f)$ is the market premium, SMB measures the historic excess returns of small market capitalization stock over big market capitalization stocks, HML is the historic excess returns of value stocks over growth stocks, the event window is (-1, +1) and the estimation window is (-150, -30)	Authors' Calculation
<i>Independent Variables</i>		
Pressure Resistant Buy(Sell)	An indicator variable that takes the value of 1 if the block buyer (seller) is an investment firm, such as, banks, mutual funds, investment advisory and brokerage firms, insurance companies, pension funds, hedge funds, private equity, foreign institutional investor, or an independent individual investor, and 0 otherwise.	Authors' categorisation
Insider Buy(Sell)	An indicator variable that takes the value of 1 if the block buyer (seller) is either a promoter, promoter company, promoter trust, executive/promoter director, and 0 otherwise.	Authors' categorisation
Pressure Sensitive Buy(Sell)	An indicator variable that takes the value of 1 if the block buyer (seller) is a corporate firm (private or public), and 0 otherwise	Authors' categorisation
Pressure Resistant Delta	An indicator variable that takes the value of 1 if the block buyer belongs to Pressure Resistant Buy category and the block seller belongs to either Pressure Sensitive or Insider category	Authors' categorisation
Insider Delta	An indicator variable that takes the value of 1 if the block buyer belongs to Insider Buy category and the block seller belongs to either Pressure Sensitive or Pressure Resistant buy category	Authors' categorisation
Pressure Sensitive Delta	An indicator variable that takes the value of 1 if the block buyer belongs to Pressure Sensitive	Authors' categorisation

	Buy category and the block seller belongs to either Pressure Resistant or Insider category	
Percentage Deal Size	Total number of shares purchased during the block trade as a percentage of the total shares outstanding on the day of the deal.	Authors' categorisation
Board Size	Total number of Directors on the board	CMIE Prowess
Board Size Piece	Defined as (Board Size minus Median Board Size)*D _B , where D _B is a dummy which takes a value 1 if the Board Size is greater than Median Board Size and 0 otherwise.	CMIE Prowess
Independent Director	Percentage of Independent Directors on the board of Directors	CMIE Prowess
Executive Director	Percentage of Executive Directors on the board of Directors	CMIE Prowess
Promoter Director	Percentage of Promoters on the board of Directors	CMIE Prowess
PB	Price to Book Ratio	CMIE Prowess
Leverage	Long Term Debt to Net-worth Ratio	CMIE Prowess
Market Cap	Logarithm of Market Capitalisation	CMIE Prowess
Regulation 2009	A dummy variable that takes the value of 1 for block trades post 2009, and value of 0 for trades between 2005—2009.	CMIE Prowess
Inst. Holding	Percentage of shares held by institutional investors in the block deal firm one quarter before the block deal	CMIE Prowess
Large Inst. Holding	A dummy variable that takes the value of 1 if the Institutional Holding is greater than 5%, or 0 otherwise	Authors' Categorisation
New Blockholder	A dummy variable that takes the value of 1 if the block holder does not have any shareholding in the one year before the block deal	Authors' Categorisation
Promoter Holding	Percentage of shares held by promoters one quarter before the block deal	CMIE Prowess
Family Owned	A dummy variable that takes the value of 1 if the block deal firm has greater than 20 percent promoter holding, else it is considered as non-family firm with value 0	Authors' Categorisation
Family Managed	A dummy variable that takes the value of 1 if the block deal firm has promoter as either a CEO or	Authors' Categorisation

Family Owned & Managed	MD or Chairman, else it is considered as non-family managed with value 0 A dummy variable that takes the value of 1 if the block deal firm has greater than 20 percent promoter holding and the promoter is either a CEO or MD or Chairman of the firm, else it is considered as non-family Owned & Managed with value 0	Authors' Categorisation
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Table 2: Descriptive Statistics

Variable	Units	Obs	Median	Mean	Std. Dev.	Min	Max
Dependent Variable							
CAR_FF_15030	Continuous	1329	0.392	1.555	11.786	-37.682	48.306
Independent Variable							
Pressure Resistant Buy	Categorical	1329	0	0.354	0.479	0	1
Pressure Sensitive Buy	Categorical	1329	0	0.236	0.424	0	1
Insider Buy	Categorical	1329	0	0.410	0.492	0	1
Pressure Resistant Sell	Categorical	1329	0	0.464	0.499	0	1
Pressure Sensitive Sell	Categorical	1329	0	0.176	0.381	0	1
Insider Sell	Categorical	1329	0	0.360	0.480	0	1
Pressure Resistant Delta	Categorical	1329	0	0.120	0.325	0	1
Pressure Sensitive Delta	Categorical	1329	0	0.184	0.387	0	1
Insider Delta	Categorical	1329	0	0.178	0.382	0	1
Percentage Deal Size Buy	Percentage	1320	0.732	2.035	4.863	0.002	79.688
Percentage Deal Size Sell	Percentage	1320	0.667	1.916	4.497	0.004	79.688
Board Size	Nos.	1329	8	7.946	4.643	0	23
Board Size Piece	Nos.	1329	0	1.807	2.738	0	15
No. of Independent Director	Nos.	1329	4	3.762	2.535	0	13
No. of Executive Director	Nos.	1329	2	2.764	2.282	0	18
No. of Promoter Director	Nos.	1329	0	1.033	1.473	0	7
Independent Director	Percentage	1329	0.474	0.413	0.231	0	1
Executive Director	Percentage	1329	0.286	0.315	0.236	0	1
Promoter Director	Percentage	1329	0	0.124	0.169	0	0.6
Market Cap	Logarithm	1329	8.077	8.143	2.830	0	14.691
PB	Ratio	1329	1.5	3.225	5.236	0	59.48
Leverage	Ratio	1329	0	0.327	1.679	-16.264	11.239
Regulation 2009	Categorical	1329	1	0.764	0.425	0	1
Family Owned	Categorical	1329	1	0.506	0.500	0	1
Family Managed	Categorical	1329	0	0.019	0.136	0	1
Family Owned Managed	Categorical	1329	0	0.339	0.474	0	1
Institutional Holding	Percentage	1329	5.9	12.624	14.581	0	59.49
Large Institutional Holding	Categorical	1329	1	0.520	0.500	0	1
New Block	Categorical	1329	1	0.712	0.453	0	1

Table 3: Market Reaction to Block Trades

Panel A: Cumulative Abnormal Returns for all Block Firms

Event Window	CAR %
(-1,+1)	1.375***
(-2,+2)	0.744*
(-3,+3)	0.05
(-5,+5)	-1.364*
(-10,+10)	-0.764

Estimation Window: (-150, -30); Number of Firm Transactions: 1065

Panel B: Cumulative Abnormal Returns around Block categories using Buy, Sell and Delta Trades

Block Category	Buy	No. of Obs.	Sell	No. of Obs.	Delta	No. of Obs.
Pressure Resistant	2.062***	471	0.796*	617	4.813***	159
Pressure Sensitive	-0.538	313	3.617***	234	(-)1.153*	244
Insider	2.32***	545	1.527***	478	4.317***	236

Estimation Window: (-150, -30)

Panel C: Cumulative Abnormal Returns around with-in group and across-group Block Trades

Buyer	Seller	Mean CAR	No. of Obs.
Pressure Resistant	Pressure Resistant	0.659	312
Pressure Sensitive	Pressure Resistant	-2.279***	161
Insider	Pressure Resistant	4.528***	144
Pressure Resistant	Pressure Sensitive	5.02***	73
Pressure Sensitive	Pressure Sensitive	1.639	69
Insider	Pressure Sensitive	3.988***	92
Pressure Resistant	Insider	4.63***	86
Pressure Sensitive	Insider	1.03	83
Insider	Insider	0.794	309

Estimation Window: (-150, -30),

Panel D: Cumulative Abnormal Returns around Institutional and Ownership Characteristics of Block Firms

Category	Mean	No. of Obs.	Category	Mean	No. of Obs.
Pre Regulation 2009	1.157**	329	Post Regulation 2009	1.145***	1082
Family Owned	2.208***	718	Non-Family Owned	0.22	693
Family Owned and Managed	1.335**	457	Non-Family Owned but Managed	1.483***	476

			Non-Family Owned and Non-Managed	-2.51**	180
Large Shareholding	0.677*	744	<5% Shareholding	1.850***	667
New Blockholder	0.889**	1020	Old Blockholder	2.124***	391
Estimation Window: (-150, -30)					

Executive Director (16)	-.08	-.15	.25	-.07	.16	-.18	-.03	-.16	.29	-.13	-.04	-.06	.24	.04	.03	1								
Market Cap (17)	-.14	.23	-.14	-.10	.18	-.20	.03	.09	-.11	-.17	-.13	-.14	.47	.48	.14	.03	1							
PB (18)	-.09	.07	.09	-.15	.09	-.12	.00	.07	.14	-.14	-.04	-.07	.05	.09	-.14	.08	.2	1						
Leverage (19)	.08	.06	-.02	-.04	.03	.15	-.15	-.02	-.03	.09	-.15	-.15	.05	.00	-.08	.01	-.04	.03	1					
Regulation 2009 (20)	.00	-.08	.05	.04	.11	-.07	-.06	-.14	.04	-.04	-.05	-.02	.23	.06	.29	.30	.13	.13	.10	1				
Family Owned (21)	.10*	.00	.06	-.05	.06	-.01	-.06	.03	.02	.06	.06	.04	.00	.06	-.05	-.12	.10	-.03	-.02	-.05	1			
Family Owned Managed (22)	-.01	-.05	-.14	.17	-.18	-.05	.23	-.07	-.13	-.08	-.01	-.01	.13	.00	.28	.20	.01	-.08	-.05	.11	-.73	1		
Large Preshare (23)	-.04	.28	-.24	-.07	.09	-.10	-.02	.05	-.20	-.04	-.09	-.10	.35	.33	.10	-.16	.53	.10	-.07	-.13	.00	.08	1	
New Block (24)	-.04	.26	.10	-.34	.24	-.02	-.24	.12	.09	-.08	.02	.01	.04	.07	-.05	.04	.03	.05	.12	-.09	.05	-.19	.04	1

Table 5: Regression analysis of the wealth effects of Buy, Sell and Delta trades, as a function of blockholder categories, while controlling for corporate governance and block's financials.

Dependent Variable: CAR

The Table shows the regression results corresponding to determinants of *CAR*. The independent variables are block categories (pressure-resistant, pressure-sensitive and insider), percentage deal size, corporate governance characteristics (board size, percent of executive and independent directors on board), and firm financials (PB, Market Cap, Leverage). We include industry and year dummies. The definition of the variables is given in Table 1. Column (1) contains buy-side block trades, column (2) contains sell-side block trades, and column (3) shows results for delta-trades. Standard Errors are in parentheses. *p<0.01, **p<0.05, ***p<0.001

Variable Name	(1)	(2)	(3)
Pressure Resistant Buy	3.751*** (0.926)		
Insider Buy	3.294*** (0.895)		
Pressure Resistant Sell		-0.900 (0.971)	
Insider Sell		-0.968 (0.973)	
Pressure Resistant Delta			3.824*** (1.024)
Insider Delta			3.205*** (0.961)
Percentage Deal Size Buy	0.0707 (0.0690)		0.0848 (0.0683)
Percentage Deal Size Sell		0.111 (0.0745)	
Board Size	0.738** (0.234)	0.649** (0.236)	0.643** (0.233)
Board Size Piece	-0.864** (0.327)	-0.751* (0.330)	-0.698* (0.327)
Executive Director	-6.711*** (1.753)	-7.432*** (1.768)	-6.487*** (1.759)
Independent Director	-4.614* (2.291)	-3.039 (2.303)	-3.155 (2.280)
Market Cap	-1.138*** (0.170)	-0.962*** (0.171)	-0.941*** (0.167)
PB	-0.0280 (0.0710)	-0.0304 (0.0713)	-0.00198 (0.0713)

Leverage	0.306 (0.220)	0.411 (0.223)	0.387 (0.218)
constant	-5.267* (2.058)	-1.367 (1.969)	-5.097* (2.044)
Year Dummy	Yes	Yes	Yes
Firm Dummy	Yes	Yes	Yes
N	1320	1320	1320
Adj. R^2	0.103	0.091	0.104

Table 6: Regression analysis of the wealth effects of block trades, as a function of blockholder categories, while incorporating the effect of regulatory intervention, and controlling for corporate governance and block's financials.

Dependent Variable: CAR

The Table shows the regression results corresponding to determinants of *CAR* while incorporating the regulatory intervention in 2009 related to Equity Listing. The independent variables are block categories (pressure-resistant, pressure-sensitive and insider), percentage deal size, corporate governance characteristics (board size, percent of executive and independent directors on board), firm financials (PB, Market Cap, Leverage) and regulatory dummy (=1 post 2009 regulation, else 0). We include industry and year dummies. In column (1), we consider the effect of regulation dummy 2009. In column (2), we conduct sub-sample analysis for firms in the pre-regulation period, and column (3) presents results for post-regulation period. Standard Errors are in parentheses. *p<0.01, **p<0.05, ***p<0.001.

	(1)	(2)	(3)
Pressure Resistant Delta	3.824*** (1.024)	2.215 (1.414)	4.411** (1.338)
Insider Delta	3.205*** (0.961)	-3.133 (1.668)	4.584*** (1.163)
Percentage Deal Size	0.0848 (0.0683)	0.175 (0.107)	0.0413 (0.0846)
Board Size	0.643** (0.233)	0.841 (0.461)	0.774** (0.291)
Board Size Piece	-0.698* (0.327)	-1.103 (0.584)	-0.848* (0.409)
Executive Director	-6.487*** (1.759)	-5.493 (5.008)	-6.121** (1.996)
Independent Director	-3.155 (2.280)	-10.03* (4.054)	-1.689 (2.740)
Market Cap	-0.941*** (0.167)	-0.408 (0.252)	-1.083*** (0.213)
PB	-0.00198 (0.0713)	-0.245* (0.111)	0.0920 (0.0937)
Leverage	0.387 (0.218)	0 (.)	0.360 (0.239)
Regulation 2009	8.303** (3.067)		
Constant	-5.097* (2.044)	0.0430 (2.048)	6.585* (2.771)
Year Dummy	Yes	Yes	Yes
Industry Dummy	Yes	Yes	Yes

N	1320	311	1009
Adj. R^2	0.104	0.116	0.111

Table 7: Regression analysis of the wealth effects of block trades, as a function of blockholder categories, while considering the effect of family-owned and family-managed firms, and controlling for corporate governance and block's financials.

Dependent Variable: CAR

The Table shows the regression results corresponding to determinants of *CAR* in the presence of family ownership and management. Columns (1), (2) and (3), presents results pertaining to family owned firms (promoter holds greater than 20% ownership). In column (1), we introduce the family owned dummy. In column (2) we present sub-sample analysis for family owned firms, and column (3) for non-family owned firms. Similarly, columns (4), (5) and (6), correspond to family owned and managed firms (promoter holds greater than 20% ownership, and the promoter is either chairman, CEO or MD of the firm). In column (4), we introduce the family owned and managed dummy. In column (5), we present the results from sub-sample analysis for family owned and managed firms, and column (6) for non-family owned and managed firms. All specifications include delta trades for block categories (pressure-resistant, pressure-sensitive and insider), percentage deal size, corporate governance characteristics (board size, percent of executive and independent directors on board), firm financials (PB, Market Cap, Leverage) and regulatory dummy (=1 post 2009 regulation, else 0). We include industry and year dummies. Standard Errors are in parentheses. *p<0.01, **p<0.05, ***p<0.001

	(1)	(2)	(3)	(4)	(5)	(6)
Pressure Resistant						
Delta	3.772*** (1.025)	3.853** (1.318)	1.605 (1.681)	3.759*** (1.026)	1.560 (2.201)	4.453*** (1.189)
Insider Delta	3.149** (0.962)	4.612*** (1.310)	-0.972 (1.658)	3.168** (0.962)	-0.433 (1.842)	4.627*** (1.201)
Percentage Deal Size						
Buy	0.0783 (0.0685)	0.0568 (0.0893)	0.170 (0.109)	0.0803 (0.0684)	0.109 (0.118)	0.0840 (0.0858)
Board Size	0.652** (0.233)	1.315*** (0.348)	0.388 (0.371)	0.659** (0.234)	0.289 (0.559)	1.092*** (0.302)
Board Size Piece	-0.723* (0.328)	-1.353** (0.483)	-0.472 (0.510)	-0.733* (0.329)	-0.426 (0.714)	-1.071* (0.421)
Executive Director	-6.204*** (1.771)	-7.682** (2.561)	-2.934 (3.046)	6.392*** (1.761)	6.845 (4.969)	-9.261*** (2.086)
Independent Director	-2.588 (2.317)	-10.95** (3.543)	1.478 (3.504)	-2.671 (2.325)	3.109 (5.962)	-6.528* (2.752)
Market Cap	-0.939*** (0.167)	1.318*** (0.261)	1.047*** (0.245)	0.940*** (0.167)	1.255*** (0.274)	-1.055*** (0.236)
PB	0.00437 (0.0714)	0.0701 (0.0907)	0.0417 (0.134)	-0.00372 (0.0713)	0.0990 (0.186)	0.0194 (0.0825)

Leverage	0.412 (0.219)	-0.585 (0.448)	1.128*** (0.290)	0.396 (0.218)	2.689** (0.907)	0.0416 (0.236)
Regulation 2009	7.340* (3.148)	- 10.67*** (2.785)	5.895 (4.653)	8.216** (3.068)	3.815 (5.228)	8.371* (3.656)
Family Owned & Managed				-0.794 (0.749)		
Family Owned	0.959 (0.711)					
Constant	-5.037* (2.044)	14.01*** (3.154)	-1.802 (2.392)	-5.028* (2.045)	4.304 (5.944)	-6.198** (2.092)
Year Dummy	Yes	Yes	Yes	Yes	Yes	Yes
Industry Dummy	Yes	Yes	Yes	Yes	Yes	Yes
N	1320	669	651	1320	446	874
Adj. R^2	0.105	0.144	0.102	0.104	0.075	0.149

Table 8: Regression analysis of the wealth effects of block trades, as a function of blockholder categories, and interaction with corporate governance characteristics.

Dependent Variable: CAR

The Table shows the regression results corresponding to determinants of *CAR* in the presence of blockholder categories and their interaction with corporate governance variables. In Column (1) we interact pressure-resistant blockholders with board size piece, percent of executive and independent directors on board. To avoid issues of multicollinearity, we use mean centered values of the interaction variables. In column (2) and (3), we present subsample of pressure-resistant and insider block firms. All specifications include other independent variables, percentage deal size, corporate governance characteristics (board size, percent of executive and independent directors on board), firm financials (PB, Market Cap, Leverage), regulatory dummy (=1 post 2009 regulation, else 0), industry and year dummies. Standard Errors are in parentheses. *p<0.01, **p<0.05, ***p<0.001

	(1)	(2)	(3)
Pressure Resistant Delta	9.306*** (2.356)		
Insider Delta	4.627*** (1.093)		
Percentage Deal Size	0.0869 (0.068)	0.384 (0.344)	-0.00335 (0.188)
Board Size	0.565* (0.235)	1.903* (0.892)	1.263 (0.824)
Board Size Piece	-0.461 (0.329)	-2.862* (1.276)	-2.113 (1.130)
Executive Director	-7.204*** (1.801)	-7.503 (7.608)	6.786 (6.019)
Independent Director	-1.008 (2.387)	-23.98** (8.037)	-23.89** (7.599)
Market Cap	-1.015*** (0.169)	-0.770 (0.913)	-1.153* (0.484)
PB	0.003 (0.071)	-0.689 (0.389)	0.312 (0.495)
Leverage	0.388 (0.217)	1.911 (1.248)	-1.122 (0.591)
Board Size*Pressure Resistant Delta	-0.232 (0.228)		
Board Size Piece*Pressure Resistant Delta	-1.052* (0.467)		
Executive Director*Pressure Resistant Delta	13.90* (5.793)		
Independent Director*Pressure Resistant Delta	-15.70*** (4.620)		

Regulation2009	10.74*** (3.108)	8.104 (10.23)	16.92* (7.362)
Constant	-7.292*** (2.136)	8.983 (5.009)	-4.386* (2.113)
Year Dummy	Yes	Yes	Yes
Industry Dummy	Yes	Yes	Yes
N	1320	158	228
Adj. R^2	0.116	0.162	0.177

Table 9: Regression analysis of wealth effects of block deals as a function of blockholder categories and ownership characteristics of the block firm, while controlling for financials and corporate governance characteristics of the block firm.

Dependent Variable: CAR

The Table shows the regression results corresponding to determinants of CAR. The independent variables are blockholder Categories: Pressure Resistant Delta, Insider Delta; Percentage Deal Size; firm financials: PB, Market Cap, Leverage; Corporate Governance Characteristics: Board Size, Executive Director, Independent Director, and Ownership Characteristics: Large Inst. holding, New blockholder, and industry and year dummies. Column (1) shows the effect of Large Inst. holding along with the financial and corporate governance variables as controls. In Column (2), we consider the case of sub sample of Large Institutional Holding. In Column (3) we consider the sub sample Non-Large Inst. holding. In Column (4), we introduce New blockholder. In Column (5), we consider a sub sample of block firms with New blockholders. In Column (6), we consider the case of block firms where the block purchase is undertaken by Old blockholders. The definition of the variables are given in Table 1. Standard Errors are in parentheses. *p<0.01, **p<0.05, ***p<0.001

	(1)	(2)	(3)	(4)	(5)	(6)
Pressure Resistant Delta	3.827*** (1.026)	1.784 (1.030)	7.218*** (1.892)	3.952*** (1.029)	3.998*** (1.134)	2.786 (2.491)
Insider Delta	3.211*** (0.965)	2.440* (1.083)	5.540** (1.714)	3.066** (0.967)	1.441 (1.248)	3.623* (1.620)
Percentage Deal Size Buy	0.0841 (0.0692)	-0.155 (0.136)	-0.0300 (0.0945)	0.0908 (0.0684)	0.0888 (0.0749)	-0.189 (0.199)
Board Size	0.647** (0.242)	0.490 (0.299)	0.190 (0.407)	0.647** (0.233)	0.273 (0.285)	1.038* (0.518)
Board Size Piece	-0.703* (0.335)	-0.889* (0.380)	0.557 (0.627)	-0.698* (0.327)	-0.407 (0.388)	-1.113 (0.759)
Executive Director	-6.517*** (1.812)	3.508 (2.396)	-8.698** (3.014)	-6.393*** (1.760)	-3.842 (2.121)	-8.525* (3.672)
Independent Director	-3.160 (2.282)	-9.515*** (2.661)	-1.545 (3.834)	-3.180 (2.279)	0.452 (2.741)	-9.732* (4.791)
Market Cap	-0.936*** (0.181)	-0.674*** (0.171)	-1.793*** (0.411)	-0.930*** (0.168)	-0.832*** (0.189)	-1.081** (0.403)
PB	-0.00199 (0.0713)	-0.0315 (0.0753)	0.0645 (0.127)	-0.000286 (0.0713)	-0.0147 (0.0814)	-0.0797 (0.160)
Leverage	0.386 (0.218)	-0.400 (0.232)	1.688** (0.526)	0.416 (0.219)	1.163*** (0.285)	-1.120* (0.526)
Large Institutional Holding	-0.0599 (0.874)					
New Block				-0.963 (0.739)		

Regulation 2009	8.279** (3.087)	-3.644 (2.302)	24.65*** (5.514)	7.661* (3.105)	2.705 (3.646)	3.738 (3.823)
Constant	-5.103* (2.046)	5.146* (2.055)	-8.083** (2.641)	-4.061 (2.193)	-3.875 (2.142)	9.852* (4.147)
Year Dummy						
Industry Dummy	1320	688	632	1320	945	375
N	1320	688	632	1320	945	375
Adj. R^2	0.103	0.157	0.157	0.105	0.138	0.122